Wind Farm Suitability Analysis—Halver, Germany

GIS Certification Capstone Project - GIS-Akademie Dortmund - December 2025

Group project (4 members) – My contributions: ArcPy scripting, cartographic design, report writing

Project Overview

This project used GIS-based multi-criteria suitability analysis to identify potential wind farm sites in the municipality of Halver, Märkischer Kreis, North Rhine-Westphalia (NRW). The analysis integrated official cadastral, topographic, and land-use datasets, applied spatial exclusion criteria (hard and soft constraints), and produced a scored suitability index for each candidate area, visualised as a map series in ArcGIS Pro.

Category	Tools & Methods
GIS Software	ArcGIS Pro, ArcPy (Python scripting interface)
Spatial Analysis	Multi-criteria suitability analysis, buffer analysis, union, spatial join, pairwise clip
Data Sources	ALKIS/ATKIS official cadastral & topographic data, Geoportal NRW, Geofabrik OpenStreetMap
Cartography	ArcGIS Pro map series, traffic light scoring visualisation, aerial basemap
Programming	Python (ArcPy)—automated multi-step spatial workflow with buffering, filtering, scoring, and geometry repair

My Contribution: ArcPy Automation Script

The analytical core of the project was a Python script using ArcPy that automated the entire multi-step spatial workflow—from data import and classification through buffering, union, scoring, and geometry repair. The script processed all input layers through a structured nine-step process:

Steps 1–3: Environment setup, study area definition (SQL query on Halver municipality boundary), and classification of input layers from a control CSV file into hard exclusion zones, soft constraint zones, and permitted areas.

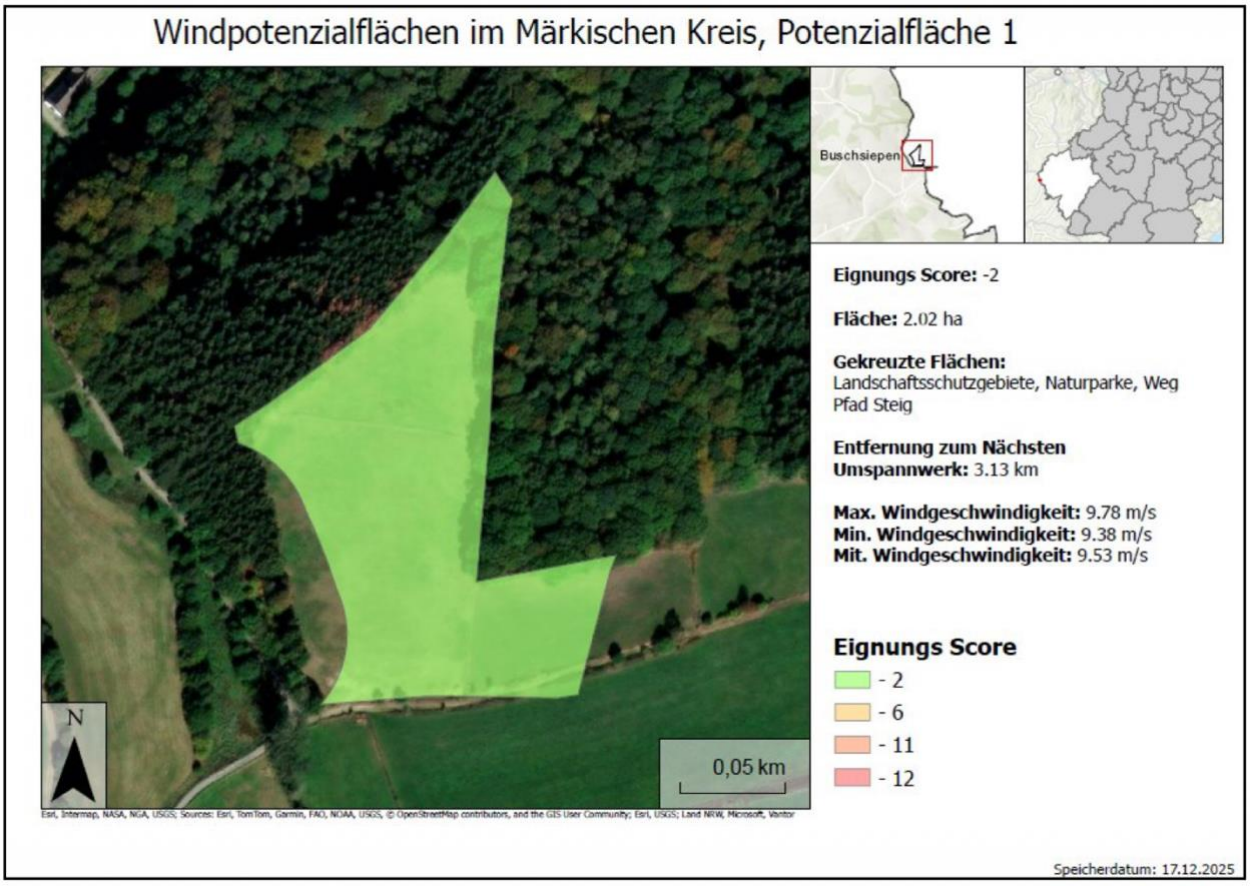
Steps 4–6: Pairwise buffering of all constraint layers, clipping to the study area, geometry repair, and union of all layers to compute all spatial overlay combinations.

Steps 7–9: SQL filtering to remove hard exclusion zones, scoring analysis to calculate cumulative suitability scores per area, and automated ArcGIS Pro map series generation for all seven identified potential sites.

After automated processing, I performed manual refinement in ArcGIS Pro: dissolving fragmented adjacent areas using the Dissolve tool, reattaching suitability scores via Spatial Join (using minimum score for conservative assessment), and adding wind speed statistics and nearest substation distances via Near analysis.

Results

Seven potential wind farm sites larger than 1 ha were identified. Sites were scored on a scale from -2 (least constrained) to -12 (most constrained) based on the number and severity of overlapping soft constraint zones. A traffic light cartographic scheme was applied for intuitive visual communication: green (-2), yellow (-6), orange/red (-11 to -12).



Site 1 — Score: -2 (best rated, 2.02 ha)

Sites 4, 5, 6 — Turbine spacing analysis

Key Findings

Site 1 was identified as the most viable location with a score of -2, 2.02 ha, minimal constraints (landscape protection zone and footpaths only), wind speeds averaging 9.53 m/s, and 3.13 km to the nearest substation.

Sites 4, 5, and 6 were spatially proximate. Applying NRW turbine spacing standards (5x rotor diameter in prevailing wind direction; ~750m given the LANUK model's 149m rotor diameter), only sites 5 and 6 could be developed simultaneously; site 4 would need to remain undeveloped.

Relevance to planning: The analysis demonstrates how automated GIS workflows can efficiently screen large areas for regulatory compliance and generate decision-ready cartographic outputs for spatial planning processes.